

```
#include <stdio.h>
```

```
int main() {
```

```
    int n, m;
```

```
    printf("Enter number of processes: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter number of resources: ");
```

```
    scanf("%d", &m);
```

```
    int alloc[n][m], max[n][m], need[n][m], avail[m];
```

```
    int finish[n], safeSeq[n];
```

```
    printf("Enter Allocation Matrix:\n");
```

```
    for (int i = 0; i < n; i++)
```

```
        for (int j = 0; j < m; j++)
```

```
            scanf("%d", &alloc[i][j]);
```

```
    printf("Enter Maximum Demand Matrix:\n");
```

```
    for (int i = 0; i < n; i++)
```

```
        for (int j = 0; j < m; j++) {
```

```
            scanf("%d", &max[i][j]);
```

```
            need[i][j] = max[i][j] - alloc[i][j];
```

```
        }
```

```
    printf("Enter Available Resources:\n");
```

```
    for (int j = 0; j < m; j++)
```

```
        scanf("%d", &avail[j]);
```

```
    for (int i = 0; i < n; i++) finish[i] = 0;
```

```
    int count = 0;
```

```

while (count < n) {
    int found = 0;
    for (int i = 0; i < n; i++) {
        if (!finish[i]) {
            int ok = 1;
            for (int j = 0; j < m; j++)
                if (need[i][j] > avail[j]) { ok = 0; break; }
            if (ok) {
                for (int j = 0; j < m; j++)
                    avail[j] += alloc[i][j];
                safeSeq[count++] = i;
                finish[i] = 1;
                found = 1;
            }
        }
    }
    if (!found) break;
}

if (count == n) {
    printf("System is in a safe state.\nSafe sequence is: ");
    for (int i = 0; i < n; i++)
        printf("P%d%s", safeSeq[i], i < n-1 ? " -> " : "\n");
} else {
    printf("System is NOT in a safe state.\n");
}

return 0;
}

```

OUTPUT :

```
Enter number of processes: 5
Enter number of resources: 3
Enter Allocation Matrix:
0 1 0
2 0 0
3 0 2
2 1 1
0 0 2
Enter Maximum Demand Matrix:
7 5 3
3 2 2
9 0 2
2 2 2
4 3 3
Enter Available Resources:
3 3 2
System is in a safe state.
Safe sequence is: P1 -> P3 -> P4 -> P0 -> P2

Process returned 0 (0x0)   execution time : 1836.928 s
Press any key to continue.
```